

Email: l.demonte@ed.ac.ukGoogle Scholar: [Publications](#)GitHub: [Repositories](#)

RESEARCH INTERESTS

- Extreme value theory, Spatial and multivariate statistics, Probabilistic forecasting
- Machine learning, generative modelling
- Environmental applications in the context of climate change.

EDUCATION

PhD Statistics, University of Edinburgh, United Kingdom Sep. 2022 – Jun. 2026

Supervisors: Dr. Ioannis Papastathopoulos, Prof. Gabriele C. Hegerl

Thesis: *Geometric considerations for generative modelling of multivariate extremes*

Outcome: Successfully defended on June 29, 2026, with no corrections required.

Visiting PhD student, KAUST, Saudi Arabia Feb. 2024 – May 2024 and Nov. 2025

Supervisor: Prof. Raphaël Huser

MSc Mathematics and Statistics, McGill University, Canada Sep. 2020 – June 2022

Supervisor: Prof. Christian Genest

Thesis: *A spatial Gamma-Gamma model for the statistical postprocessing of ensemble weather forecasts*

BSc Mathematics and Computer Science, McGill University, Canada Sep. 2017 – May 2020

AWARDS

School of Mathematics, University of Edinburgh Sep. 2023

- Full PhD Studentship: £27,600 tuition and £20,800 stipend per year

Mitacs Canada Jan. 2022

- Funding for research on extreme value modeling in partnership with Hydro-Québec. \$30,000 (CAD)

RESEARCH COLLABORATIONS AND PROFESSIONAL EXPERIENCE

Hydro-Québec Research Institute (IREQ) Jan. 2022 – Aug. 2022

- Research partnership to develop methods for the statistical post-processing of weather forecasts, specifically of extreme meteorological events.

DataHub, National Bank of Canada May 2019 – Sept. 2020

Data Science Internship

- Development and automation of the official DataHub credit card database.
- Statistical modeling of credit card use and behaviours.

TEACHING AND MENTORING

Teaching Assistantships (University of Edinburgh)

Tutoring workshops. Marking of assignments and examinations. List of courses:

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| - MATH10093 – Statistical Computing | Jan. – May 2023, Jan. – May 2025, and Jan. – May 2026 |
| - MATH08065 – Computing and Numerics | Jan. – May 2026 |
| - MATH10096 – Applied Statistics | Jan. – May 2025 and Jan. – May 2026 |
| - MATH11199 – Python Programming | Sep. – Dec. 2025 |
| - MATH08066 – Probability | Sep. – Dec. 2023 and Sep. – Dec. 2025 |
| - MATH10064 – Multivariate Data Analysis | Jan. – May 2025 |
| - MATH11176 – Extended Statistical Programming | Sep. – Dec. 2023 |
| - MATH11187 – Generalised Regression Models | Sep. – Dec. 2023 |
| - MATH08051 – Statistics | Jan. – May 2023 |

StatsBase Statistics Helpdesk (University of Edinburgh) Nov. 2025

Providing help to MSc students in Statistics for the Statistical Methodology and Theory of Statistical Inference courses.

Teaching Assistantships (McGill University)

Preparation of tutorial sessions, office hours for course-related questions. Marking of midterm and final exams, supervision of assignment marking. List of courses:

- MATH 324 – Statistics Jan. 2022 – May 2022
- MATH 323 – Probability Sep. 2021 – Dec. 2021
- MATH 203 – Principles of Statistics Jan. 2021 – May 2021

Statistics Helpdesk (McGill University)

Sep. 2020 – Dec. 2021

Providing help to undergraduate students in Statistics and Probability. Theoretical and programming guidance for courses in the range MATH 2xx–5xx.

Undergraduate Project Supervision (McGill University)

Guidance of three students towards the completion of an undergraduate project in extreme value theory.

- MATH 470 – Honours Research Project Jan. 2022 – May 2022
- MATH 410 – Majors Project Jan. 2021 – May 2021 and Sep. 2021 – Dec. 2021

ACADEMIC SERVICE AND LEADERSHIP

Postgraduate representative, School of Mathematics (University of Edinburgh)

Representative of the postgraduate students of the *School of Mathematics* Sept. 2024 – Oct. 2025

- Intermediary between the students and the School administration.
- Organisation of social events within the School.
- Representing the *School of Mathematics* at the *Martingale PhD Navigator Programme* events for future scholars (December 2024 and 2025).

Scientific article reviewing

Journal of Statistics and Computing.

PRESENTATIONS AND POSTER SESSIONS

Invited

- Generative AI Modelling for Extreme Events (GAME), Bologna, Italy: *Multivariate extremal index and clustering in multivariate extremes: A case study of Dutch hydrological extremes.* Jun. 2026
- CfS Annual Conference 2026, Edinburgh, United Kingdom: *Multivariate extremal index and clustering in multivariate extremes: A case study of Dutch hydrological extremes.* Jun. 2026
- Generative AI Modelling for Extreme Events, University of Edinburgh Generative AI Laboratory, Edinburgh, United Kingdom: *Generative modelling of multivariate geometric extremes using normalising flows.* Jun. 2025
- Statistics seminar, University of Exeter, Exeter, United Kingdom: *Geometric extreme value theory – A normalising flows approach.* Feb. 2025
- CMStatistics, London, United Kingdom: *Geometric extreme value theory – A normalising flows approach.* Dec. 2024
- CMStatistics, Berlin, Germany: *Bayesian inference for radially-stable distributions.* Dec. 2023
- BIRS-IMAG Modern Statistical and Machine Learning Approaches for High-Dimensional Compound Spatial Extremes, Granada, Spain (short presentation): *Geometric inference for Hüsler–Reiss random vectors.* May 2023
- Séminaires de sciences de la décision HEC Montréal, Montréal, Canada: *Multivariate extremes – A geometric Bayesian inference approach.* Sep. 2023

Contributed

- 14th International Conference on Extreme Value Analysis, University of North Carolina Chapel Hill, United States: *Generative modelling of multivariate geometric extremes using normalising flows* Jun. 2025
- 15th International Meeting on Statistical Climatology, Toulouse, France: *Flood risk modelling using geometric extreme value theory.* Jun. 2024
- CfS Annual Conference 2024, Edinburgh, United Kingdom (poster): *Multivariate radial Pareto distributions: a geometric approach to the statistical modelling of multivariate extremes.* Jun. 2024
- 13th International Conference on Extreme Value Analysis, Milan, Italy (poster): *Bayesian approach to geometric inference for multivariate extremes.* Jun. 2023

SOFTWARE (OPEN-SOURCE)

- Python code [geometricExtremesNF](#): Creator and maintainer. Based on De Monte et al. (2025)
- R package [geometricExtremes](#): Creator and maintainer. Based on Papastathopoulos et al. (2023).
- R code [TSExtremes](#): Collaborator. Based on Auld et al. (2026).

TECHNICAL SKILLS

Languages: R, Python, Julia, Bash, SQL, LaTeX.

High-performance computing: Eddie (University of Edinburgh), Ibex (King Abdullah University of Science and Technology), CASIR (Hydro-Québec).

Cloud computing: Microsoft Azure (National Bank of Canada).

LANGUAGES

Bilingual: English, French.

PUBLICATIONS

Research

- De Monte, L. & Papastathopoulos, I. (2026), *A directional Poisson point process for univariate extremes*, [Statistics & Probability Letters](#) **235**, 110722.
- De Monte, L., Huser, R., Papastathopoulos, I., and Richards, J. (2025) *Generative modelling of multivariate geometric extremes using normalising flows*, [arXiv](#) preprint.
- Papastathopoulos, I., De Monte, L., Campbell, R. & Rue, H. (2023) *Statistical inference for radially-stable generalised pareto distributions and return level-sets in geometric extremes*, [arXiv](#) preprint.

Book chapter

- Auld, G., De Monte, L. & Papastathopoulos, I. (2026) Time series in extremes. In de Carvalho, M., Huser, R., Naveau, P., and Reich, B. J., editors, *Handbook on Statistics of Extremes*. Chapman & Hall/CRC, Boca Raton, FL.

Other

- Richards, J., De Monte, L. Review of “Risk Revealed: Cautionary Tales, Understanding and Communication” by Paul Embrechts, Marius Hofert, and Valérie Chavez-Demoulin. [JABES](#) (2024).
- Genest, C. and De Monte, L. (2021) *À la recherche de la Rondurie*, [Accromath](#), Volume 16 p.24-29.

Theses

- De Monte, L. *Geometric considerations for generative modelling of multivariate extremes*. PhD thesis. [The University of Edinburgh](#), 2026.
- De Monte, L. *A Spatial Gamma-Gamma Model for the Statistical Postprocessing of Ensemble Weather Forecasts*. Master’s thesis. [McGill University](#), 2022.